



# Dong-Uk Seo

ROBOTICS ENGINEER, SLAM

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## Summary

Ph.D. candidate researching the robotics field in KAIST Urban Robotics Lab. Research interests lie in LiDAR-Camera fusion-based SLAM for robust long-term autonomy. Additionally interested in the process collaborated with multi-agent and localization in the dense map representation.

## Education

### KAIST(Korea Advanced Institute of Science and Technology)

PH. D. CANDIDATE IN DEPARTMENT OF ELECTRICAL ENGINEERING, URBAN ROBOTICS LAB

- GPA: 4.16 / 4.30

Daejeon, South Korea

Mar. 2023 - Present

### KAIST(Korea Advanced Institute of Science and Technology)

M.S. IN DEPARTMENT OF ELECTRICAL ENGINEERING, URBAN ROBOTICS LAB

- GPA: 4.17 / 4.30

Daejeon, South Korea

Mar. 2021 - Feb. 2023

### Yonsei University

B.S. IN DEPARTMENT OF ELECTRICAL & ELECTRONIC ENGINEERING

- GPA: 4.06 / 4.30
- National Science and Engineering Excellence Scholarship (2016 - 2019, Korea Scholarship Foundation)
- Undergraduate Research Assistant in Compiler Research Laboratory (CORELAB)
- Graduation Thesis: A Deep Learning Based Super-Resolution RF Map Reconstruction Scheme for Fingerprinting Localization

Seoul, South Korea

Mar. 2014 - Aug. 2020

## Experiences & Projects

### RGB/Event/IMU fusion based SLAM with photo-realistic mapping for dense reconstruction.

KAIST URBAN ROBOTICS LAB & ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE (ETRI)

- Developed RGB/event/IMU fusion-based SLAM for robust pose estimation across challenging illumination and high-speed motion conditions.
- Applied radiance field based algorithms for photo-realistic dense spatial map reconstruction.
- Participated as a **team leader**.

Daejeon, South Korea

June. 2025 - Present

### Visual SLAM for indoor autonomous navigation

KAIST URBAN ROBOTICS LAB & SAMSUNG ELECTRONICS

- Visual SLAM is performed indoors using a camera, IMU, and wheel encoder
- Participate as a **team leader**.

Daejeon, South Korea

June. 2024 - May. 2025

### Multi-robot collaboration for dense map reconstruction and localization

KAIST URBAN ROBOTICS LAB & YONSEI UNIVERSITY & KOREA UNIVERSITY & NAVER LABS & INCHEON AIRPORT

- While removing dynamic objects, multi-robots collaborate to localize and map with a deep learning-based algorithm for reconstructing a dense representation.
- Participate as a team member.

Daejeon, South Korea

July. 2023 - Present

### Robust localization system in a challenging environment for last-mile delivery.

KAIST URBAN ROBOTICS LAB & MOBINN

- An indoor/outdoor day/night localization system for an obstacle-overcoming delivery robot is researched. LiDAR-Visual-Inertial fusion-based SLAM frameworks with deep learning technology are implemented.
- Participate as a **team leader**.

Daejeon, South Korea

May. 2023 - Sep. 2024

## Monocular RGB-D Camera and IMU sensor based Navigation Framework on a mobile robot

Daejeon, South Korea

KAIST URBAN ROBOTICS LAB & HYUNDAI KEFICO

April. 2022 - Jan. 2023

- Implemented Visual-Inertial SLAM using monocular RGB-D camera and IMU sensor. The global planner and local planner are also implemented on the map refined by Surfel, and finally the robot could proceed with autonomous navigation toward the destination.
- Participated as a **team leader**.

## Autonomous Driving SLAM framework (Technology transfer)

Daejeon, South Korea

KAIST URBAN ROBOTICS LAB & HILLSLOGIS

Dec. 2021 - Mar. 2023

- Relatively inexpensive sensors such as 2D LiDAR, RGB-D camera, and wheel are used to perform accurate localization and map generation, helping mobile robots to automatically run processes in a factory-like environment. In this project, camera-LiDAR-wheel odometry fusion-based simultaneous localization and mapping (SLAM) was performed.
- **Awarded CES 2023 Innovation Award**
- Participated as a team member.

## Visual SLAM on mobile smartphones for AR application

Daejeon, South Korea

KAIST URBAN ROBOTICS LAB & ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE (ETRI)

Mar. 2021 - Present

- For AR application, SLAM is performed on smartphones. To be more general applications, only monocular cameras and IMU sensors have been used, while point and line features are utilized. Also develop an algorithm to recognize planes.
- Participate as a team member.

## Last-mile project

Daejeon, South Korea

KAIST URBAN ROBOTICS LAB & KOREA ELECTRONICS TECHNOLOGY INSTITUTE (KETI)

Nov. 2020 - Dec. 2021

- Development of self-driving robot technology capable of learning one's own driving experience and moving to the destination robustly in four seasons, day and night, using various sensor information for driving in indoor and outdoor urban living environments and using uncertain maps.
- Participated as a team member.

## Development of big data intelligent information system

Seoul, South Korea

KOREA DATA AGENCY (KDA) & KOREA UNIVERSITY

June. 2019 - Sep. 2019

- Symptom-based Dentist Matching System using BiLSTM-attention.

## Publications

\* These authors contributed equally

## International

### Robust Structureless Monocular Visual-Inertial Initialization Exploiting Line Features and Vanishing Points

Junwan Choi, Woongrae Jo, **Dong-Uk Seo**, Jinwoo Jeon, Hyun Myung  
2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

2026

### PLED-VINS: A Point-Line Event-Based Visual Inertial SLAM for Dynamic Environments

Seunghun Lee\*, Jihun Nam\*, **Dong-Uk Seo**, Hyun Myung  
2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

2026

### GaussianFlow SLAM: Monocular Gaussian Splatting SLAM Guided by GaussianFlow

**Dong-Uk Seo**, Jinwoo Jeon, Eungchang Mason Lee, Hyun Myung  
IEEE Robotics and Automation Letters (RA-L) 11.6 (2026) pp. 7644–7651. IEEE

2026

### AIM-SLAM: Dense Monocular SLAM via Adaptive and Informative Multi-View Keyframe Prioritization with Foundation Model

Jinwoo Jeon, **Dong-Uk Seo**, Eungchang Mason Lee, Hyun Myung  
2026 IEEE International Conference on Robotics and Automation (ICRA)

2026

### PanoNetVLAD: Visual loop closure detection in continuous space represented with panoramic view using multiple cameras

Sungjae Shin, Yeeun Kim, Byeongho Yu, Eungchang Mason Lee, **Dong-Uk Seo**, Hyun Myung  
2023 The 23rd International Conference on Control, Automation and Systems (ICCAS)

2023

## Enhancing Robustness of Line Tracking Through Semi-Dense Epipolar Search in Line-based SLAM

**Dong-Uk Seo**, Hyungtae Lim, EungChang Mason Lee, Hyunjun Lim, Hyun Myung

2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

2023

## Struct-MDC: Mesh-refined unsupervised depth completion leveraging structural regularities from visual SLAM

Jinwoo Jeon, Hyunjun Lim, **Dong-Uk Seo**, Hyun Myung

IEEE Robotics and Automation Letters (RA-L) 7.3 (2022) pp. 6391–6398. IEEE

2022

## PaGO-LOAM: Robust ground-optimized LiDAR odometry

**Dong-Uk Seo\***, Hyungtae Lim\*, Seungjae Lee, Hyun Myung

2022 19th International Conference on Ubiquitous Robots (UR)

2022

## QR-SCAN: Traversable Region Scan for Quadruped Robot Exploration using Lightweight Precomputed Trajectory

EungChang Mason Lee, **DongUk Seo**, JinWoo Jeon, Hyun Myung

2021 21st International Conference on Control, Automation and Systems (ICCAS)

2021

## Domestic

### Hybrid Feature Tracking Method for Pose Estimation in Dynamic Environments

**Dong-Uk Seo**, Dasol Hong, Hyun Myung

Korea Robotics Society Annual Conference (KRoC 2025)

2025

### Surfel-based Depth Image Update for Dynamic Object-Removed Map Generation

**Dong-Uk Seo**, Sungjae Shin, Hyun Myung

Korea Robotics Society Annual Conference (KRoC 2023)

2023

### Optimal Path Planning System using Candidates from Road-Based Sliding Window

**Dong-Uk Seo**, Hyun Myung

Institute of Control, Robotics and Systems (ICROS) (2022) pp. 421–422

. 2022

## Honors & Awards

### Hilti × Trimble 360 Visual-Inertial SLAM Challenge — 1st Place, SLAM

#### Category

Vienna, Austria

IEEE ICRA 2026

2026

- Jinwoo Jeon, **Dong-Uk Seo**, Junwan Choi, Seunghun Lee, Jihun Nam, Hyungtae Lim, and Hyun Myung